

Problem 1 (15 pts)

- Write a program to determine if a given string is a palindrome
 - Input: you will be given an input file with one string per line
 - Output: for each line output if the string is a palindrome
 - Output: the total number of palindromes
 - E.g.
 - Input: geek
asleelsa
thisisatesttsetasisiht
 - Output: false
true
true
2

Problem 2 (20 pts)

- Using Recursion
 - Create a function that prints and counts the unique set of sub-strings of a given string
 - Input: abcab
 - Output: bca, bc, abc, cab, c, b, a, abca, ca, ab, abcab, bcab -> 12

Problem 3 (20 pts)

- Using Recursion
 - Write a function that reverses a stack implemented as an array
 - Write a function that reverses a stack implemented as a Linked-list
 - Write a function that reverses a stack implemented as a doubly linked-list
- Input: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10
- Output: 10, 9, 8, 7, 6, 5, 4, 3, 2, 1

Problem 4 (45 pts)

- A group of n Ghostbusters is battling n ghosts. Each Ghostbuster carries a proton pack which shoots a stream at a ghost destroying it. A stream goes in a straight line and continues when it hits the ghost. The Ghostbusters decide upon the following strategy:
 - They will pair off with the ghosts, forming n Ghostbuster-ghost pairs and then simultaneously each Ghostbuster will shoot a stream at their chosen ghost. The streams are not allowed to cross so each Ghostbuster must choose a pairing for which no streams will cross.
 - You can assume that there are an equal number Ghostbusters and ghosts.
- Create a solution which will determine if all of the Ghosts can be eliminated in a single shot of the Ghostbusters by returning the string "All Ghosts: " <were|were not> " eliminated"
- Partial credit will be given for the design and algorithm and full credit will be given for a correct working solution.
- I will provide a set test cases in the format:
 - Line 1 will contain the number of Ghostbusters
 - Lines 2- n will contain a ghostbuster, ghost pairing represented as a series of letters and numbers space separated with a B indicating the position of a Ghostbuster, a G indicating the position of a ghost. The information following either a B or a G will be the X Y coordinate of either the ghostbuster or the ghost.

Problem 4 (con't)

- A sample input file might look like:

2
B 0 0 G 5 5
B 3 0 G 0 3

